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Parameterization of Seed Functions for Equivalent Representations of Time-Varying Delay Systems

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***Abstract*— Abel's classic transformation shows that any well-posed system with time-varying delay is equivalent to a parameter-varying system with fixed delay. The existence of such a parameter-varying constant delay representation then simplifies the problems of stability analysis and optimal control. Unfortunately, the method for construction of such transformations has been ad-hoc – requiring an iterative time-stepping approach to constructing the transformation beginning with a seed function subject to boundary-value constraints. Moreover, a poor choice of seed function often results in a constant delay representati on with large time-variations in system parameters – obviating the benefits of such a representation. In this paper, we show how the set of all feasible seed functions can be parameterized using a basis for L_2. This parameterization is then used to search for seed functions for which the corresponding time-transformation results in smaller parameter variation. The parameterization of admissible seed functions is illustrated with numerical examples that contrast how well-chosen and poorly chosen seed functions affect the boundedness of a time transformation.**

I. Introduction

Time-varying delays appear in networked control, teleoperation, and biological regulation, where even small time-variations can alter stability and performance. However, analysis and control of systems with time-varying delay is complicated by the typically associated time-varying notion of state-space [1]–[4]. To address this issue, a classical remedy, dating back to Abel-type transformations, shows that any well-posed system with a time-varying delay can be recast as a parameter-varying system with a fixed delay; with a change of time variables, one regains a time-invariant state space [5]–[11]. In the fixed-delay representation, standard tools— Linear Matrix Inequalities (LMIs), Integral Quadratic Constraints (IQCs), and the Partial Integral Equations (PIEs) framework, can then be used for analysis and control design [12]–[15].

The conversion of a system with time varying delay to parameter-varying system with fixed delay is based on a given time-transformation $t \mapsto h(t)$ which must satisfy certain recursive constraints (Lem. 1). However, this time-transformation is not unique, but rather is parameterized by a choice of *seed function*, ϕ which specifies the transformation on an initial time interval. This seed function must itself satisfy certain boundary constraints as determined by the time-varying delay. Then, for any valid seed function, the corresponding time-transformation may be constructed iteratively.

Unfortunately, however, as is demonstrated in Section V, not all seed functions result in well-behaved time-transformations. Specifically, we show that many choices of seed function result in a time-transformation whose derivative, $\dot{h}$ grows over time. This growth in $\dot{h}$ translates directly into large parameter variations in the resulting fixed-delay representation – an attempt to quantify this growth for systems with periodic time-varying delays via perturbative expansion can be found in [16]. This parameter variation then prevents the use of techniques such as normalization [11], [17] to the fixed-delay representation.

The goal of this paper, then, is to propose a parameterization of seed functions which then allows one to find time-transformations with desirable properties such as periodicity and boundedness. Specifically, we provide an L_2 parameterization of the set of admissible seed functions (S_τ), defining a affine operator $\mathcal{T} : L_2 \to S$ such that for any $\nu \in L_2$, $\phi = \mathcal{T}\nu \in S$. Furthermore, we show that for any $\phi \in S$, $\mathcal{T}\phi''' = \phi$ – establishing a one-to-one map from L_2 to S. By working directly with the seed parameters, ν, rather than the seed itself, one can enforce constraints on the seed functions (and resulting time-transformations). For example, using a polynomial basis for L_2, one can use Sum of Squares (SoS) [18] to enforce non-negativity of ϕ'.

To establish the map from L_2 to seed functions, ϕ, to time-transformation, h, to constant delay representation, we begin in Section III by defining the class of admissible time-transformations and showing that for any such time-transformation, solutions of the resulting fixed-delay and time-varying delay systems are equivalent – a slight extension of classical results in [5]–[7]. We then define the set of admissible seed functions and show how any such seed function defines a resulting time-transformation, h.

In Section IV, we then establish the invertible mapping between seed parameters and seed functions – also showing that the well-known quadratic seed function is a special case and that monotonicity of the seed function can be enforced using SoS constraints on the seed parameter.

Finally, in Section V, we motivate the proposed parameterization by demonstrating the impact of a seed function choice on the derivative of the time transformation, $\dot{h}$ — comparing the effect of quadratic, exponential, and affine plus sinusoidal seed functions on $\dot{h}$ for a time-varying sinusoidal delay.

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II. Notation

$\mathbb{N}^n$, $\mathbb{R}^n$, and $\mathbb{R}^n_+$ denote the space of $n-$dimensional vectors of natural, real, and positive real numbers, respectively. $L_2([a,b],\mathbb{R})$ is the space of square–integrable functions; $\mathcal{H}^3([a,b],\mathbb{R})$ is the Sobolev space of functions with third (weak) derivative in L_2. Composition is written $(g\circ f)(x) = g(f(x))$. Moreover, $f^{\circ k}$ denotes k-fold composition, and $(f^{-1})^{\circ k}$ the k-fold composition of the inverse (when it exists). For compact $\Omega \subset \mathbb{R}$, $\mathcal{C}^k(\Omega)$ denotes the space of k-times continuously differentiable functions $f : \Omega \to \mathbb{R}$ with norm $\|f\| = \sup_{t\in\Omega} \|f(t)\|_2$.

III. Equivalence Between Variable- and Fixed-Delay DDEs

Consider a delay differential equation (DDE) of the form:

$$\begin{aligned} \dot{x}(t) &= A_0 x(t) + A_1 x\big(t-\tau(t)\big), \quad t \geq 0, \\ x(t) &= \zeta(t), \quad t \in [-\tau(0),\, 0], \end{aligned} \tag{1}$$

where $A_0,\ A_1 \in \mathbb{R}^{n_x\times n_x}$. We assume the initial function $\zeta \in \mathcal{C}([-\tau(0),0],\mathbb{R}^{n_x})$ is continuous, and the time-varying delay $\tau \in \mathcal{C}^1(\mathbb{R},\mathbb{R}_+)$ is bounded with $\dot{\tau}(t) < 1$ for all $t \geq 0$.

A. Time Transformations

A *time transformation*, h, may be used to convert a DDE with time varying delay to a parameter-varying DDE with fixed time delay. This function is invertible and changes the time variable t to a new time variable defined as $\lambda := h^{-1}(t)$. The time transformation $h(\lambda)$, must be a strictly increasing function constructed to satisfy an Abel equation. This Abel equation ensures that the delay $\tau(t)$ at time $t = h(\lambda)$ aligns with a constant shift τ^* in the new time λ. This is formalized in the following lemma which is a slight modification of [17, Thm. 1].

Lemma 1. *Suppose $\tau^* \in \mathbb{R}_+$ and $\tau \in \mathcal{C}^1(\mathbb{R},\mathbb{R}_+)$ with $\dot{\tau}(t) < 1$. Let $h \in \mathcal{C}^1([-\tau^*,\infty),[-\tau(0),\infty))$ be a strictly increasing unbounded function, with $h(0) = 0$ and where*

$$h(\lambda) - \tau(h(\lambda)) = h(\lambda - \tau^*), \quad \forall \lambda \geq 0. \tag{2}$$

For any $\zeta \in \mathcal{C}[-\tau(0),0]$, if $x(t)$ satisfies

$$\begin{aligned} \dot{x}(t) &= A_0 x(t) + A_1 x(t-\tau(t)), \quad \forall t \geq 0 \\ x(t) &= \zeta(t), \quad \forall t \in [\,-\tau(0),\,0\,] \end{aligned} \tag{3}$$

then $\bar{x}(\lambda) = x(h(\lambda))$ and $\bar{\zeta} = \zeta\big(h(\lambda)\big)$ satisfy

$$\begin{aligned} \dot{\bar{x}}(\lambda) &= \dot{h}(\lambda)A_0\bar{x}(\lambda) + \dot{h}(\lambda)A_1\bar{x}(\lambda-\tau^*), \quad \forall \lambda \geq 0 \\ \bar{x}(\lambda) &= \bar{\zeta}\big(\lambda\big), \quad \forall \lambda \in [-\tau^*,0]. \end{aligned} \tag{4}$$

Conversely, if $\bar{\zeta} \in \mathcal{C}[-\tau^*,0]$, and $\bar{x}$ satisfies Eqn. (4)*, then $x(t) = \bar{x}(h^{-1}(t))$ and $\zeta\big(t\big) = \bar{\zeta}(h^{-1}(t))$ satisfy Eqn.* (3)*.*

Proof: By the inverse function theorem, $h^{-1} \in \mathcal{C}^1$ and

$$(h^{-1})'(t) = \frac{1}{\dot{h}\big(h^{-1}(t)\big)}, \qquad t \in [-\tau(0),\infty).$$

(⇒) Let $\zeta \in \mathcal{C}([-\tau(0),0])$ and suppose x solves (3). Define $\bar{x}(\lambda) := x\big(h(\lambda)\big)$. For $\lambda \geq 0$,

$$\begin{aligned} \dot{\bar{x}}(\lambda) &= \dot{x}\big(h(\lambda)\big)\dot{h}(\lambda) \\ &= \dot{h}(\lambda)\Big[A_0 x\big(h(\lambda)\big) + A_1 x\big(h(\lambda) - \tau(h(\lambda))\big)\Big]. \end{aligned}$$

Using (2), we have: $\dot{\bar{x}}(\lambda) = \dot{h}(\lambda)\Big[A_0\bar{x}(\lambda) + A_1\bar{x}(\lambda-\tau^*)\Big]$. For the initial segment, if $\lambda \in [-\tau^*,0]$ then $h(\lambda) \in [h(-\tau^*),h(0)] = [-\tau(0),0]$, hence $\bar{x}(\lambda) = x\big(h(\lambda)\big) = \zeta\big(h(\lambda)\big)$. Thus (4) holds.

(⇐) Let $\bar{\zeta} \in \mathcal{C}([-\tau^*,0])$ and suppose $\bar{x}$ satisfies (4). Define $x(t) := \bar{x}\big(h^{-1}(t)\big)$ for $t \geq 0$. Then

$$\begin{aligned} \dot{x}(t) &= \dot{\bar{x}}(h^{-1}(t))\,(h^{-1})'(t) = \dot{\bar{x}}(h^{-1}(t))\,\frac{1}{\dot{h}(h^{-1}(t))} \\ &= A_0\bar{x}(h^{-1}(t)) + A_1\bar{x}(h^{-1}(t)-\tau^*) \end{aligned}$$

From (2) , we obtain

$$h^{-1}\big(t-\tau(t)\big) = h^{-1}(t) - \tau^*,$$

so $\bar{x}(h^{-1}(t)-\tau^*) = \bar{x}\big(h^{-1}(t-\tau(t))\big) = x\big(t-\tau(t)\big)$. Hence $\dot{x}(t) = A_0x(t) + A_1x\big(t-\tau(t)\big)$, which is the differential equation in (3). For the initial segment, if $t \in [-\tau(0),0]$ then for $h^{-1}(t) \in [-\tau^*,0]$, and thus for $\zeta(t) = \bar{\zeta}(h^{-1}(t))$,

$$x(t) = \bar{x}\big(h^{-1}(t)\big) = \bar{\zeta}\big(h^{-1}(t)\big) =: \zeta(t),$$

so (3) holds. ∎

Lemma 1 establishes equivalence of solutions between a system with time-varying delay and a fixed-delay system with parameter-varying uncertainty. We note, however, that the choice of time-transformation, h, may not be uniquely defined. Nor have we suggested any approach to constructing such a transformation. In the following subsection, we show that such time-transformations may be parameterized through the choice of seed function, ϕ.

B. Recursive Construction of the Time Transformation

Lemma 1 defines a map from a set of admissible time-transformations to a set of equivalent fixed-delay representations of a system with time-varying delay. In this subsection, we show how such time-transformations may be parameterized by choice of admissible seed function, ϕ.

The set of admissible seed functions is defined next based on [17, Thm. 1], [11, Sec. 2.3].

Definition 1. *We say that ϕ is a admissible seed function associated to $\tau^*, \tau(0) > 0, \dot{\tau}(0) < 1$ if ϕ is strictly increasing and $\phi \in S_{\dot{\tau}(0),\tau(0),\tau^*}$ where*

$$S_{\dot{\tau}_0,\tau_0,\tau^*} := \tag{5}$$
$$\left\{\phi \in \mathcal{H}^3[-\tau^*,0] \,:\, \begin{matrix}\phi(0) = 0,\\ \phi(-\tau^*) = -\tau_0,\end{matrix}\ \dot{\phi}(0) = \frac{\dot{\phi}(-\tau^*)}{1-\dot{\tau}_0}\right\}$$

For any given admissible seed function, we may construct an associated time transformation h using the recursion defined in [11], [17]. Specifically, given $\tau(t)$ and $\phi \in S_{\dot{\tau}(0),\tau(0),\tau^*}$, let $h \in \mathcal{C}^1([-\tau^*,\infty),\mathbb{R}_+)$ be defined as:

$$h(\lambda) := \begin{cases} \phi(\lambda), & \lambda \in [-\tau^*,0] \\ \theta^{-1}\big(h(\lambda-\tau^*)\big), & \lambda \geq 0 \end{cases} \tag{6}$$

where $\theta(t) := t - \tau(t)$ is strictly increasing and unbounded and hence is invertible. Equivalently, for $k \in \mathbb{N}$,

$$h(\lambda) := (\theta^{-1})^{\circ k}\big(\phi(\lambda - k\tau^*)\big) \qquad \lambda \in [(k-1)\tau^*, k\tau^*]. \quad (7)$$

Moreover, for $k \in \mathbb{N}$ and $\lambda \in [(k-1)\tau^*, k\tau^*]$,

$$\dot{h}(\lambda) = \frac{\dot{\phi}(\lambda - k\tau^*)}{\prod_{n=1}^{k} \dot{\theta}\big((\theta^{-1})^{\circ(k-n+1)}(\phi(\lambda - n\tau^*))\big)}. \quad (8)$$

As shown in [11], [17], for any admissible seed function, this recursion results in a time-transformation, h which satisfies the conditions of Lemma 1. However, the seed function for a given time-varying delay is not uniquely defined. It is restricted only by the values of $\tau(0)$ and $\dot{\tau}(0)$ (and the choice of τ^*). Furthermore, we note that the corresponding fixed delay representation in Eqn. (4) includes the multiplicative time-varying parameter, $\dot{h}(\lambda)$. From Eqn. (8) we see that $\dot{h}(\lambda)$ will depend on the choice of seed function. Thus, while all seed functions will result in equivalent system representations, some seed functions may result in fixed-delay representations with large or unbounded parameter variation. In the following section, we consider a parameterization of seed functions which will allow us to search for seed functions which result in time transformations with certain desirable properties.

IV. Parameterization of Seed Functions

In this section, we provide an invertible affine map ($\mathcal{T}$) between L_2 and the set of seed functions $S_{\tau_0',\tau_0,\tau^*}$. The key observation here is that any seed function, ϕ, is uniquely determined by its third derivative, ϕ'''. Following the main result in Thm. 2, we provide SoS conditions for ϕ to be increasing and show that the previously used quadratic seed function is a special case of the proposed parameterization.

Define operator $\mathcal{T}$ as follows.

$$(\mathcal{T}\nu)(\lambda) = -\tau_0 + \tau_0\beta(\lambda) + \int_{-\tau^*}^{0} K(\lambda,\eta)\nu(\eta)\,d\eta + \int_{-\tau^*}^{\lambda} \frac{(\lambda-\eta)^2}{2}\nu(\eta)\,d\eta, \quad (9)$$

where

$$\beta(\lambda) = \frac{2(1-\tau_0')}{(2-\tau_0')\tau^*}(\lambda+\tau^*) + \frac{\tau_0'}{(2-\tau_0')(\tau^*)^2}(\lambda+\tau^*)^2, \quad (10)$$

and

$$K(\lambda,\eta) = \frac{1-\tau_0'}{2-\tau_0'}\left(-\eta - \frac{\eta^2}{\tau^*}\right)(\lambda+\tau^*) - \frac{1}{2-\tau_0'}\left(\frac{\tau_0'\eta^2}{2(\tau^*)^2} - \frac{1-\tau_0'}{\tau^*}\eta\right)(\lambda+\tau^*)^2. \quad (11)$$

The following theorem shows that $\mathcal{T}$ defines an invertible map from L_2 to $S_{\tau_0',\tau_0,\tau^*}$.

Theorem 2. *Given τ^*, $\tau_0 > 0$ and $\tau_0' < 1$. Let $S_{\tau_0',\tau_0,\tau^*}$ be defined as in Eqn.* (5) *and $\mathcal{T}$ be defined as in Eqn.* (9)*. Then:*

1) *For any $\phi \in S_{\tau_0',\tau_0,\tau^*}$, $\mathcal{T}\phi''' = \phi$.*
2) *For any $\nu \in L_2[-\tau^*, 0]$, $\mathcal{T}\nu \in S_{\tau_0',\tau_0,\tau^*}$.*
3) *For any $\nu \in L_2[-\tau^*, 0]$, $(\mathcal{T}\nu)''' = \nu$.*

Proof: For 1), since $\phi \in S_{\tau_0',\tau_0,\tau^*} \subset \mathcal{H}^3[-\tau^*, 0] \subset \mathcal{C}^2[-\tau^*, 0]$, we apply the Taylor formula with integral reminder [19]: for any $f \in \mathcal{H}^{k+1}([a,b])$ and for any $x \in [a,b]$,

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x-a)^k + \int_a^x f^{(k+1)}(\xi)\,\frac{(x-\xi)^k}{k!}\,d\xi. \quad (12)$$

Applying (12) with $f = \phi$, $a = -\tau^*$ and $k = 2$ yeilds

$$\phi'(0) = \phi'(-\tau^*) + \tau^*\phi''(-\tau^*) - \int_{-\tau^*}^{0} \eta\phi'''(\eta)\,d\eta, \quad (13)$$

$$\phi(0) = \phi(-\tau^*) + \tau^*\phi'(-\tau^*) + \frac{1}{2}(\tau^*)^2\phi''(-\tau^*) + \int_{-\tau^*}^{0} \frac{\eta^2}{2}\phi'''(\eta)\,d\eta, \quad \text{and} \quad (14)$$

$$\phi(\lambda) = \phi(-\tau^*) + (\lambda+\tau^*)\phi'(-\tau^*) + \frac{1}{2}(\lambda+\tau^*)^2\phi''(-\tau^*) + \int_{-\tau^*}^{\lambda} \frac{(\lambda-\eta)^2}{2}\phi'''(\eta)\,d\eta. \quad (15)$$

$\phi \in S_{\tau_0',\tau_0,\tau^*}$ implies $\phi'(0) = \frac{\phi'(-\tau^*)}{(1-\dot{\tau}_0)}$, which combined with (13) yields

$$\phi''(-\tau^*) = \frac{1}{\tau^*}\left[\frac{\tau_0'}{1-\tau_0'}\phi'(-\tau^*) + \int_{-\tau^*}^{0} \eta\phi'''(\eta)\,d\eta\right]. \quad (16)$$

Likewise combining $\phi(0) = 0$ and $\phi(-\tau^*) = -\tau_0$ with (14) imply

$$\phi''(-\tau^*) = \frac{2}{(\tau^*)^2}\left[\tau_0 - \tau^*\phi'(-\tau^*) - \int_{-\tau^*}^{0} \frac{\eta^2}{2}\phi'''(\eta)\,d\eta\right]. \quad (17)$$

Substituting (17) into (13) and rearranging yields

$$\phi'(-\tau^*) = \frac{1-\tau_0'}{2-\tau_0'}\left[\frac{2\tau_0}{\tau^*} - \frac{1}{\tau^*}\int_{-\tau^*}^{0} \eta^2\phi'''(\eta)\,d\eta - \int_{-\tau^*}^{0} \eta\phi'''(\eta)\,d\eta\right]. \quad (18)$$

Substituting (18) into (16), we get:

$$\begin{aligned}\phi''(-\tau^*) &= \frac{1}{\tau^*}\left[\frac{\tau_0'}{2-\tau_0'}\left(\frac{2\tau_0}{\tau^*} - \frac{1}{\tau^*}\int_{-\tau^*}^{0} \eta^2\phi'''(\eta)\,d\eta - \int_{-\tau^*}^{0} \eta\phi'''(\eta)\,d\eta\right) + \int_{-\tau^*}^{0} \eta\phi'''(\eta)\,d\eta\right] \\ &= \frac{2\tau_0'}{2-\tau_0'}\frac{\tau_0}{(\tau^*)^2} - \frac{\tau_0'}{2-\tau_0'}\frac{1}{(\tau^*)^2}\int_{-\tau^*}^{0} \eta^2\phi'''(\eta)\,d\eta + \frac{2(1-\tau_0')}{2-\tau_0'}\frac{1}{\tau^*}\int_{-\tau^*}^{0} \eta\phi'''(\eta)\,d\eta. \quad (19)\end{aligned}$$

Now, substituting (18) and (19) into (15) and using the

boundary condition $\phi(-\tau^*) = -\tau_0$, we obtain:

$$\begin{aligned}\phi(\lambda) &= -\tau_0 + (\lambda+\tau^*)\frac{1-\tau_0'}{2-\tau_0'}\left[\frac{2\tau_0}{\tau^*}\right.\\ &\qquad \left. -\frac{1}{\tau^*}\int_{-\tau^*}^{0}\eta^2\phi'''(\eta)\,d\eta - \int_{-\tau^*}^{0}\eta\phi'''(\eta)\,d\eta\right]\\ &\quad + \frac{(\lambda+\tau^*)^2}{2}\left[\frac{2\tau_0'}{2-\tau_0'}\frac{\tau_0}{(\tau^*)^2}\right.\\ &\qquad -\frac{\tau_0'}{2-\tau_0'}\frac{1}{(\tau^*)^2}\int_{-\tau^*}^{0}\eta^2\phi'''(\eta)\,d\eta\\ &\qquad \left. +\frac{2(1-\tau_0')}{2-\tau_0'}\frac{1}{\tau^*}\int_{-\tau^*}^{0}\eta\phi'''(\eta)\,d\eta\right]\\ &\quad + \int_{-\tau^*}^{\lambda}\frac{(\lambda-\eta)^2}{2}\phi'''(\eta)\,d\eta\\ &= -\tau_0 + \tau_0\beta(\lambda)\\ &\quad + \int_{-\tau^*}^{0}K(\lambda,\eta)\phi'''(\eta)\,d\eta + \int_{-\tau^*}^{\lambda}\frac{(\lambda-\eta)^2}{2}\phi'''(\eta)\,d\eta\\ &= (\mathcal{T}\phi''')(\lambda), \quad \text{for all } \lambda\in[-\tau^*,0].\end{aligned}$$

Next, we establish 3) by showing that for any $\nu \in L_2[-\tau^*,0]$, if $\phi = \mathcal{T}\nu$, then $\phi''' = \nu$. Recall from (9) that

$$\begin{aligned}(\mathcal{T}\nu)(\lambda) &= -\tau_0 + \tau_0\beta(\lambda)\\ &\quad + \int_{-\tau^*}^{0}K(\lambda,\eta)\nu(\eta)\,d\eta + \int_{-\tau^*}^{\lambda}\frac{(\lambda-\eta)^2}{2}\nu(\eta)\,d\eta.\end{aligned}$$

From the definitions of β in (10) and K in (11) , we observe that $\beta(\lambda)$ and $K(\lambda,\eta)$ are at most quadratic in λ. Hence, $\beta'''(\lambda) = 0$ and $\partial_\lambda^3 K(\lambda,\eta) = 0$. Next, we note that

$$\begin{aligned}&\frac{d}{d\lambda}\int_{-\tau^*}^{\lambda}\frac{(\lambda-\eta)^2}{2}\nu(\eta)\,d\eta = \int_{-\tau^*}^{\lambda}(\lambda-\eta)\nu(\eta)\,d\eta,\\ \Rightarrow\quad &\frac{d^2}{d\lambda^2}\int_{-\tau^*}^{\lambda}\frac{(\lambda-\eta)^2}{2}\nu(\eta)\,d\eta = \int_{-\tau^*}^{\lambda}\nu(\eta)\,d\eta,\\ \Rightarrow\quad &\frac{d^3}{d\lambda^3}\int_{-\tau^*}^{\lambda}\frac{(\lambda-\eta)^2}{2}\nu(\eta)\,d\eta = \nu(\lambda).\end{aligned}$$

Thus, we conclude that $(\mathcal{T}\nu)''' = \nu$.

For 2), suppose $\nu \in L_2[-\tau^*,0]$. As per Eqn. (5), we need to show that $\mathcal{T}\phi \in H^3$, $(\mathcal{T}\phi)(0) = 0$, $(\mathcal{T}\phi)(-\tau^*) = -\tau_0$, and $(\mathcal{T}\phi)'(0)(1-\dot\tau) = (\mathcal{T}\phi)'(-\tau^*)$. First, 3) implies $\phi = \mathcal{T}\nu \in H^3$. Next, since $\beta(-\tau^*) = 0$ and $K(-\tau^*,\eta) = 0$, we have $(\mathcal{T}\nu)(-\tau^*) = -\tau_0$. Third, since $\beta(0) = 1$ and $K(0,\eta) = -\frac{\eta^2}{2}$, we have

$$(\mathcal{T}\nu)(0) = -\tau_0+\tau_0+\int_{-\tau^*}^{0}\frac{-\eta^2}{2}\nu(\eta)\,d\eta+\int_{-\tau^*}^{0}\frac{\eta^2}{2}\nu(\eta)\,d\eta = 0.$$

For the final condition, differentiating (9), we obtain:

$$\begin{aligned}(\mathcal{T}\nu)'(\lambda) &= \tau_0\beta'(\lambda) + \int_{-\tau^*}^{0}\partial_\lambda K(\lambda,\eta)\nu(\eta)\,d\eta\\ &\quad + \int_{-\tau^*}^{\lambda}(\lambda-\eta)\nu(\eta)\,d\eta, \quad \forall\lambda\in[-\tau^*,0], \quad (20)\end{aligned}$$

$$\text{where}\quad \beta'(\lambda) = \frac{2\big(1-\tau_0'\big)}{\big(2-\tau_0'\big)\,\tau^*} + \frac{2\,\tau_0'}{\big(2-\tau_0'\big)\,(\tau^*)^2}\,(\lambda+\tau^*)$$

$$\text{and}\quad \partial_\lambda K(\lambda,\eta) = \frac{1-\tau_0'}{2-\tau_0'}\left(-\eta-\frac{\eta^2}{\tau^*}\right) - \frac{2}{2-\tau_0'}\left(\frac{\tau_0'\eta^2}{2(\tau^*)^2} - \frac{1-\tau_0'}{\tau^*}\eta\right)(\lambda+\tau^*).$$

We also note that

$$\beta'(-\tau^*) = \frac{2(1-\tau_0')}{(2-\tau_0')\tau^*}$$

and

$$\partial_\lambda K(-\tau^*,\eta) = \frac{1-\tau_0'}{2-\tau_0'}\Big(-\eta-\frac{\eta^2}{\tau^*}\Big)$$

which implies

$$\begin{aligned}(\mathcal{T}\nu)'(-\tau^*) &= \tau_0\beta'(-\tau^*) + \int_{-\tau^*}^{0}\partial_\lambda K(-\tau^*,\eta)\nu(\eta)\,d\eta\\ &= \frac{1-\tau_0'}{2-\tau_0'}\left[\frac{2\tau_0}{\tau^*} - \frac{1}{\tau^*}\int_{-\tau^*}^{0}\eta^2\nu(\eta)\,d\eta - \int_{-\tau^*}^{0}\eta\nu(\eta)\,d\eta\right].\end{aligned} \quad (21)$$

Similarly, since

$$\beta'(0) = \frac{2}{(2-\tau_0')\tau^*}$$

and

$$\partial_\lambda K(0,\eta) = \frac{1}{2-\tau_0'}\Big((1-\tau_0')\eta - \frac{\eta^2}{\tau^*}\Big),$$

we have

$$\begin{aligned}(\mathcal{T}\nu)'(0) &= \tau_0\beta'(0) + \int_{-\tau^*}^{0}\partial_\lambda K(0,\eta)\nu(\eta)\,d\eta - \int_{-\tau^*}^{0}\eta\nu(\eta)\,d\eta\\ &= \frac{1}{2-\tau_0'}\left[\frac{2\tau_0}{\tau^*} - \frac{1}{\tau^*}\int_{-\tau^*}^{0}\eta^2\nu(\eta)\,d\eta - \int_{-\tau^*}^{0}\eta\nu(\eta)\,d\eta\right].\end{aligned} \quad (22)$$

Combining Eqn. (21) and (22), we obtain:

$$\frac{(\mathcal{T}\nu)'(-\tau^*)}{(\mathcal{T}\nu)'(0)} = 1-\tau_0'.$$

We conclude that $\mathcal{T}\nu \in S_{\tau_0',\tau_0,\tau^*}$.

■

Having defined a parameterization of seed functions $\mathcal{T}\nu$ for $\nu \in L_2$, we now observe that the simplest such seed function (for $\nu = 0$) is precisely the quadratic seed function derived in [17, Thm. 1].

Remark 1. *If $\nu(\eta) = 0$ for all $\eta \in [-\tau^*,0]$, then the integral terms in $\mathcal{T}\nu$ (See Eqn. (9)) vanish, and for $\lambda \in [-\tau^*,0]$,*

$$\phi(\lambda) = -\tau(0) + \tau(0)\beta(\lambda), \quad (23)$$

where $\beta(\lambda)$ in (10) is a quadratic polynomial in λ. This quadratic seed function coincides with the choice used in [17, Thm. 1].

Although Thm. 2 ensures that for any $\nu \in L_2$, $\phi = \mathcal{T}\nu \in S_{\tau_0',\tau_0,\tau^*}$, admissibility of a seed function further requires ϕ' to be non-negative. Such a condition can be imposed upon the seed parameter, ν as follows.

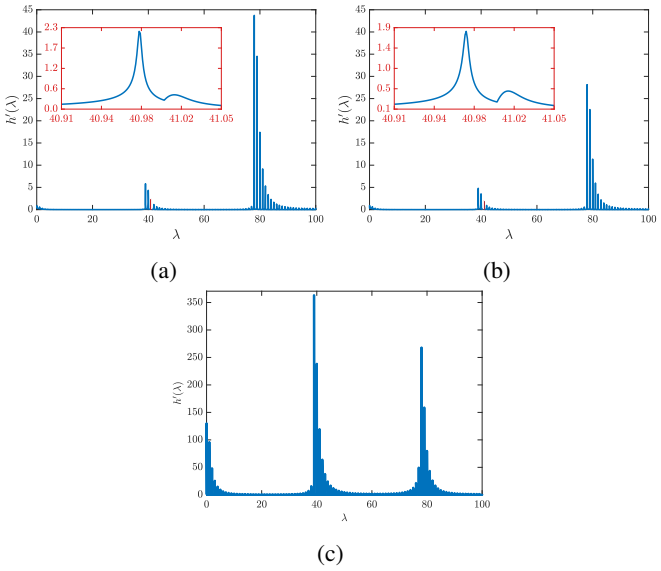


Fig. 1: The derivative $h'(\lambda)$ of the time transformation for (a) quadratic seed, (b) affine plus sinusoidal seed, and (c) exponential seed, with different maximum values of h' for horizon of 100– illustrating the impact of seed function choice on stability margins for a delay $\tau(t) = (\frac{1}{2\pi} - 0.001)\sin(2\pi t) + (\frac{1}{2\pi} + 0.001)$.

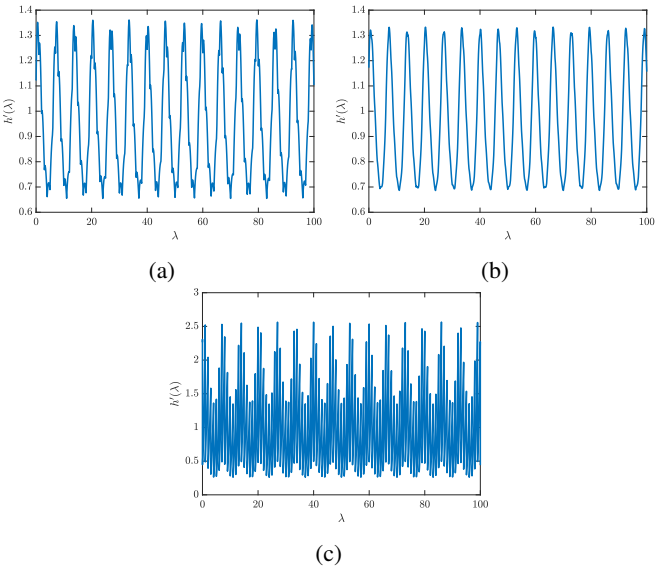


Fig. 2: The derivative $h'(\lambda)$ of the time transformation for (a) quadratic seed, (b) affine plus sinusoidal seed, and (c) exponential seed, with different maximum values of h' for horizon of 100 for the delay $\tau(t) = 1 + 0.3sin(t)$.

Corollary 2.1. *Given τ_0', τ_0, τ^*, let $\mathcal{T}$ be as in Eqn.* (9)*. For any $\nu \in L_2[-\tau^*, 0]$, $(\mathcal{T}\nu)'(\lambda) > 0$ for all $\lambda \in [-\tau^*, 0]$ if and only if*

$$\tau_0\beta'(\lambda) + \int_{-\tau^*}^{0} \partial_\lambda K(\lambda,\eta)\nu(\eta)\,d\eta + \int_{-\tau^*}^{\lambda} (\lambda-\eta)\nu(\eta)\,d\eta > 0,$$

for all $\lambda \in [-\tau^*, 0]$.

Proof: Follows from the definition of $\mathcal{T}$ in Eqn. (20). ■

If the seed parameters are polynomial, then SoS optimization can be used to enforce the non-negativity condition in Cor. 2.1. Note that since the non-negativity condition here is univariate, the corresponding univariate SoS non-negativity test is both *necessary* and *sufficient*.

V. Numerical Comparison of Seed Functions

For a given time-varying delay $\tau(t)$, the choice of seed function significantly influences the time transformation h and its derivative h', potentially complicating analysis of the fixed-delay, parameter-varying representation. In this section, we compare the effect of three choices of seed parameter on the derivative of the time-transformation, h'.

Specifically, we consider two examples of periodic time-varying delay, and for each example, examine three candidate seed parameters. For each resulting seed function, we construct $h'(\lambda)$ over an extended interval.

Specifically, recall that for $\theta(t) = t - \tau(t)$,

$$h(\lambda) := (\theta^{-1})^{\circ k}\big(\phi(\lambda - k\tau^*)\big) \qquad \lambda \in [(k-1)\tau^*, k\tau^*].$$

Because there is no analytic expression for θ^{-1}, the $h(\lambda)$ and $\dot{h}(\lambda)$ are computed pointwise in λ. That is, for every given $\lambda \in [(k-1)\tau^*, k\tau^*]$ with $k \in \mathbb{N}$, we initialize $h_0 := \phi(\lambda - k\tau^*)$ and for $j = 0, \cdots, k$, compute $h_{j+1} := (\theta^{-1})(h_j)$ where θ^{-1} is evaluated numerically using Newton iteration – yielding $h_j = (\theta^{-1})^{\circ j}\big(\phi(\lambda - k\tau^*)\big)$ and $h_k = h(\lambda)$. The h_j are then used to compute $h'(\lambda)$ as

$$h'(\lambda) = \frac{\phi'(\lambda - k\tau^*)}{\prod_{j=1}^{k} \theta'(h_j)}$$

For all examples, a time horizon of $\lambda \in [0, 100]$ is used.

a) Example 1: First consider time-varying delay $\tau(t) = \gamma_0 \sin(2\pi t) + \gamma_1$, where $\gamma_0 = \frac{1}{2\pi} - 0.001$ and $\gamma_1 = \frac{1}{2\pi} + 0.001$. Here $\tau(0) = \gamma_1$, $\tau'(0) = 2\pi\gamma_0$ and we choose $\tau^* = 1$, corresponding to the period of delay.

We now select three seed parameters $\nu \in L_2$:

$$\nu_1(\lambda) = 0, \qquad \nu_2(\lambda) = \frac{8\gamma_1}{1 - e^{-2}} e^{2\lambda}$$

$$\nu_3(\lambda) = -\Lambda_1 \left(\frac{\pi}{2}\right)^3 \sin\left(\pi(\lambda + 1)\right)$$

where

$$\Lambda_1 = \frac{4\pi\gamma_0\gamma_1}{\pi - 2\pi^2\gamma_0 + 2\gamma_1}.$$

The corresponding quadratic, exponential and affine plus sinusoidal seed functions are then

$$(\mathcal{T}\nu_1)(\lambda) = -\tau(0) + \tau(0)\beta(\lambda), \quad (\mathcal{T}\nu_2)(\lambda) = \frac{\tau(0)(e^{2\lambda} - 1)}{1 - e^{-2}}$$

and

$$(\mathcal{T}\nu)(\lambda) = \Lambda_2 - \tau(0) + \Lambda_2\lambda + \Lambda_1\left(1 - \cos\left(\frac{\pi}{2}(\lambda + 1)\right)\right),$$

where

$$\Lambda_2 = \frac{0.002\pi^2\gamma_1}{0.002\pi^2 + 2\gamma_1}$$

and where $\beta(\lambda)$ is defined in Eqn. (10) for $\tau_0 = \gamma_1$ and $\tau_0' = 2\pi\gamma_0$. It is easily verified that the seeds $\mathcal{T}(\nu_i) \in S_{\tau_0',\tau_0,\tau^*}$ are monotonic.

b) Example 2: In the second example, we use a time-varying delay $\tau(t) = 1 + 0.3\sin(t)$. Here $\tau(0) = 1$, $\tau'(0) = 0.3$ and we choose $\tau^* = 1$, which is a fraction $\frac{1}{2\pi}$ the period of delay. In this case, we again use slightly modifed version of ν_1, ν_2 and ν_3 from the first example:

$$\nu_1(\lambda) = 0, \qquad \nu_2(\lambda) = \frac{8}{1 - e^{-2}} e^{2\lambda}$$

$$\nu_3(\lambda) = -\Lambda_1 \left(\frac{\pi}{2}\right)^3 \sin\left(\pi(\lambda + 1)\right)$$

where

$$\Lambda_1 = \frac{2.6}{0.7\pi + 2}, \quad \Lambda_2 = \frac{0.6}{0.7\pi + 2}.$$

c) Analysis: The derivative of the time-transformation, $h'(\lambda)$ corresponding to the time-delays and seed parameters from Example 1 can be found in Figure 1. For Example 2, $h'(\lambda)$ is shown in Figure 2. Among the tested seed functions, the affine-plus-sinusoidal parameter produces the smallest bounds on h', followed by the quadratic function, while the exponential function results in the largest bounds. The suitability of the affine-plus-sinusoidal may be due to structural similarity between seed function and time-varying delay, suggesting the possibility of seed functions whose corresponding time-transformation admit global bounds on the derivative. These results also highlight the importance of careful selection of seed function to minimize parameter variation, thereby enhancing stability analysis in small-gain or IQC frameworks.

VI. Conclusion

Systems with time-varying delay can be equivalently represented by systems with fixed delay and multiplicative parameter variation. However, this representation is not unique, being parameterized by a time-transformation which is, in turn, defined by a seed function. The choice of seed function has a significant effect on the parameter variation of the resulting constant-delay representation. In this paper, we show that the set of admissible seed functions can be parameterized by the set of seed parameters consisting of the space of L_2 functions. We then examine several choices of seed parameter and show how this choice impacts the parameter variation of the resulting fixed-delay representation. This parameterization may allow for direct optimization of seed functions to minimize parameter variation for use in small gain or IQC type stability conditions.

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